

Learner: Aidan Elliot

Student ID: 041080471

Course: PHY4000 - Black Holes, Big Bangs and the Cosmos

Assignment: Final Project

Instructor: Dr. Asghar Gill

Due date: August 14, 2026

Repository: github.com/aidan-elliott/transit-hunter

Supplementary website: transit-hunter.aidanelliott.com

Reducing false positives in TESS transit-candidate vetting

A two-stage BLS, convolutional neural network, and diagnostic-feature pipeline

Abstract

The transit method has made it possible to search large numbers of stars for the small, periodic decrease in brightness caused when a planet crosses in front of its host star. The same surveys also produce astrophysical false positives and instrumental false alarms that can resemble planetary transits. This project tested whether a second diagnostic stage could reject more false positives than a transit-shape classifier alone. A frozen NASA Exoplanet Archive snapshot supplied labels for TESS Objects of Interest, while SPOC 120-second light curves supplied the photometry. After availability and eligibility checks, the dataset contained 1,812 labelled signals from 1,691 TESS Input Catalog targets. Whole targets were assigned to chronological training, validation, and sealed test partitions to limit target leakage and test performance on newer catalogue records. Stage 1 was a two-branch one-dimensional convolutional neural network trained on global and local phase-folded light-curve views. Stage 2 combined the Stage 1 score with BLS and vetting diagnostics such as odd-even depth difference, secondary-eclipse depth, transit symmetry, and sector-to-sector depth variation. On the sealed test set, Stage 2 increased PR-AUC from 0.664 to 0.719 and reduced the false-positive rate from 0.604 to 0.486. This was a reduction of 17 accepted false positives, with a TIC-grouped bootstrap difference of -0.118 and a 95% interval from -0.184 to -0.053. The improvement came with a loss of 19 true positives and a recall decrease from 0.851 to 0.716. The result supports diagnostic second-stage vetting, but it also shows why model thresholds must be selected according to the scientific cost of missed planets. The system ranks known candidates; it does not confirm or statistically validate new planets.

Explanatory statement connecting the project to PHY4000

This project connects directly to PHY4000 because it begins with one of the course's central ideas: almost everything astronomers know about distant objects has to be inferred from light. The Week 4 material explains how light carries information, while the Week 6 telescope material identifies time monitoring and light curves as basic astronomical measurements (Gill, 2026a, 2026b). TESS applies those ideas to exoplanets by measuring repeated changes in stellar

brightness, the transit method discussed in Week 10 (Gill, 2026c). The project then deals with a major limitation of that method, since eclipsing stars, blended sources, stellar variability, and instrumental effects can produce convincing transit-like signals. Reducing those false positives matters beyond the machine-learning exercise. Candidate reliability and survey completeness affect the populations astronomers infer from a catalogue, including how common planetary systems may be and how many worlds could eventually be considered in discussions of habitability. That connects the technical results to the Week 13 distinction between a potentially habitable environment and evidence for life, and it reinforces the need to treat uncertain observational evidence carefully (Gill, 2026d).

1. Introduction

Most extrasolar planets cannot be separated visually from their host stars. The contrast is too large and the angular separation is too small for ordinary imaging, so astronomers often detect planets indirectly. One of the most productive indirect methods is transit photometry. When an orbit is oriented correctly, a planet passes in front of its star and blocks a small fraction of the measured light. Repeated dips at a consistent period can reveal the planet's orbital period and relative size. TESS was designed to apply this method across most of the sky, with an emphasis on bright nearby stars that are suitable for later study (Ricker et al., 2015).

However, a periodic dip is not automatically evidence of a planet. Eclipsing binary stars, background binaries blended into the photometric aperture, stellar variability, and instrumental artifacts can all produce transit-like patterns. Simulations conducted before the TESS launch predicted a substantial astrophysical false-positive population alongside the expected planet detections (Sullivan et al., 2015). The practical problem is therefore split into two tasks: finding transit-like signals and deciding which signals deserve expensive follow-up. The second task is usually called candidate vetting.

As TESS accumulates observations, manual vetting alone becomes difficult to scale. Convolutional neural networks can learn shape information from phase-folded light curves, and earlier work showed that global and local light-curve views contain complementary evidence (Shallue & Vanderburg, 2018; Yu et al., 2019). A global view shows the full orbital phase, including possible secondary events, while a local view gives the primary transit more resolution. More elaborate systems such as ExoMiner combine learned representations with diagnostics used by astronomers (Valizadegan et al., 2022). ExoMiner++ later extended that approach to two-minute TESS data with additional diagnostic inputs (Valizadegan et al., 2025). These studies make a strong case for machine-assisted vetting, but they also show that a classifier's score depends on its training population, input products, and label policy. A score cannot be detached from the experiment that produced it.

Against that background, this project asked a narrow question: on a fixed, labelled sample of TESS two-minute SPOC light curves, does a second diagnostic model reduce false-positive acceptance beyond a global/local CNN while retaining useful recall? The hypothesis was that odd-even differences, secondary-eclipse evidence, transit symmetry, BLS strength, and sector consistency would add information that transit shape alone could miss. The comparison was

conducted at thresholds chosen on validation data to target 0.90 recall. A newer temporal test set was then evaluated without threshold retuning.

The result did not support a simple win for Stage 2. It rejected more false positives and improved both ROC-AUC and PR-AUC, but its validation operating point did not transfer cleanly to the newer test population. Recall fell more than expected. The second stage improved discrimination without producing a better result on every measure. Diagnostic vetting is useful here only when its threshold and error costs are stated plainly. Nothing in the experiment supports using it to confirm planets.

2. Light curves, transits, and false positives

The physical starting point is light. PHY4000 describes light as electromagnetic radiation that can be measured as an energy flow and interpreted through its interaction with matter (Gill, 2026a). Transit photometry uses a comparatively simple part of that information. Instead of separating light into a spectrum, it measures how the total detected brightness changes with time. The Week 6 material defines a light curve as a series of brightness measurements made over a period, which is exactly the data structure used in this project (Gill, 2026b).

Building on that measurement, the approximate fractional depth of an uncomplicated planetary transit is

$$\delta \approx (R_p / R_\star)^2$$

where R_p is the planetary radius and $R_\star$ is the stellar radius. The equation explains why smaller planets are harder to detect and why the same planet produces a shallower signal around a larger star. It also shows why the light curve alone does not give a complete planetary description. Stellar radius, orbital geometry, limb darkening, dilution by neighbouring sources, and noise all affect the observed event. The transit method is powerful because it converts a very small brightness change into evidence about an orbit, but the inference depends on both the instrument and the astrophysical setting (Gill, 2026c; Ricker et al., 2015).

This dependence becomes clearer in phase folding. If a trial period is close to the true period, observations from multiple orbits can be placed on a common phase axis. Repeated transit points then accumulate near the same phase, raising the signal above uncorrelated noise. The Box Least Squares algorithm searches for this type of periodic, box-shaped decrease in unevenly sampled photometric time series (Kovács et al., 2002). BLS is well matched to transit searches because a short dimming event is not sinusoidal. In this project, BLS supplied the candidate period, epoch, duration, depth, power, and signal-to-noise estimate used for later processing.

Yet a strong BLS peak can still be misleading. An eclipsing binary may produce a deep primary eclipse and a weaker secondary eclipse. If the assumed period is half of the binary's true orbital period, alternate events can have different depths. This motivates an odd-even depth comparison. A secondary eclipse near phase 0.5 is another warning sign, while an asymmetric or V-shaped event may be less consistent with a small planet crossing a stellar disk. Instrumental trends and

stellar activity create another class of difficulty because detrending can either reveal a weak signal or distort it. None of these diagnostics is decisive by itself.

For that reason, modern vetting systems combine several kinds of evidence. Shallue and Vanderburg (2018) demonstrated that a CNN could use global and local phase-folded views to rank Kepler transit signals. Yu et al. (2019) adapted deep-learning triage and vetting to TESS data, while ExoMiner explicitly incorporated diagnostic inputs that resemble the checks performed by domain experts (Valizadegan et al., 2022). These papers informed the design of this project, but their published scores are not treated as direct benchmarks. They used different missions or TESS sectors, different label definitions, larger diagnostic sets, and different train-test designs. Comparing their reported AUC values directly with this project would suggest a common test population that does not exist.

That comparison also requires a firm distinction between vetting and validation. In this report, vetting means estimating whether a known labelled signal resembles the confirmed-planet class or the false-positive/false-alarm class. Statistical validation is a stronger claim that requires a calibrated false-positive analysis, astrophysical scenario modelling, and evidence about contamination or nearby stars. Confirmation usually requires still stronger observational evidence. The present model was not designed to meet either standard. Its output is a candidate-ranking score conditioned on the frozen dataset.

3. Data and provenance

The label source was the NASA Exoplanet Archive's `toi` table. The archive documents the TESS Follow-up Observing Program Working Group dispositions used here: CP for confirmed planet, FP for false positive, FA for false alarm, PC for planet candidate, APC for ambiguous planetary candidate, and KP for known planet (NASA Exoplanet Archive, 2025). The table is a living catalogue that changes as observations and follow-up information accumulate. For reproducibility, the project saved the exact query, raw response, retrieval date, and SHA-256 checksum rather than treating the current web table as a fixed source.

Following that policy, the catalogue was retrieved on August 13, 2026. It contained 8,113 source TOI rows before the supervised label rules were applied. Confirmed planets were assigned label 1. False positives and false alarms were assigned label 0. Unresolved PC and APC records were excluded because treating them as either class would create label noise by assumption. KP records were kept separately because their discovery histories differ from TESS-confirmed objects. The resulting supervised snapshot contained 2,099 TOIs across 1,976 TIC targets. This use of TOI dispositions follows the catalogue's role as a record of candidates and follow-up outcomes, as described by Guerrero et al. (2021).

Next, the project audited whether each target had a SPOC 120-second light-curve product. SPOC produces calibrated photometric and transit-search products that are archived at the Mikulski Archive for Space Telescopes (National Aeronautics and Space Administration, n.d.; Space Telescope Science Institute, n.d.). Coverage was available for 1,834 supervised TOIs, while 265 had no matching two-minute product. A further 22 covered rows lacked a valid catalogue period required by the predeclared eligibility audit. Although catalogue ephemerides were used during

eligibility checking, neither the catalogue period nor transit midpoint entered the formal BLS search or either model. The final eligible population contained 1,812 TOIs across 1,691 TIC targets.

The acquisition process downloaded target-specific light curves and cached them outside version control because the raw FITS products are too large for the public repository. The first acquisition pass recorded 124 transient network failures. All 124 succeeded during the manifested retry, so the final model-ready dataset contained 1,812 samples and no failed samples. Each saved sample contained a 201-bin global view, a 101-bin local view, the diagnostic feature vector, its TOI and TIC identifiers, split assignment, and label. The dataset manifest also confirmed that every stored model input was finite.

The temporal split was performed only after eligibility was fixed. A TIC's most recent TOI creation date assigned the entire target group to one partition, preventing multiple signals from the same star from crossing partitions. Older groups formed the training set, later groups formed validation, and the newest groups formed the sealed test set. Table 1 reports the result. The project audit found zero TIC crossings.

Table 1 *Frozen TIC-grouped temporal partitions*

Partition	TOIs	Unique TICs	CP	FP/FA	Group-date range
Training	1,240	1,183	471	769	2018-09-05 to 2021-06-23
Validation	287	254	113	174	2021-06-23 to 2021-12-21
Sealed test	285	254	141	144	2021-12-21 to 2025-10-16
Total	1,812	1,691	725	1,087	2018-09-05 to 2025-10-16

The timeline in Figure 1 shows that the catalogue is not temporally uniform. Confirmed planets and negative dispositions arrived in different concentrations, and the latest records are comparatively sparse. This matters because the test set is not a random sample from the same time period as training. It is intentionally a modest test of temporal transfer.

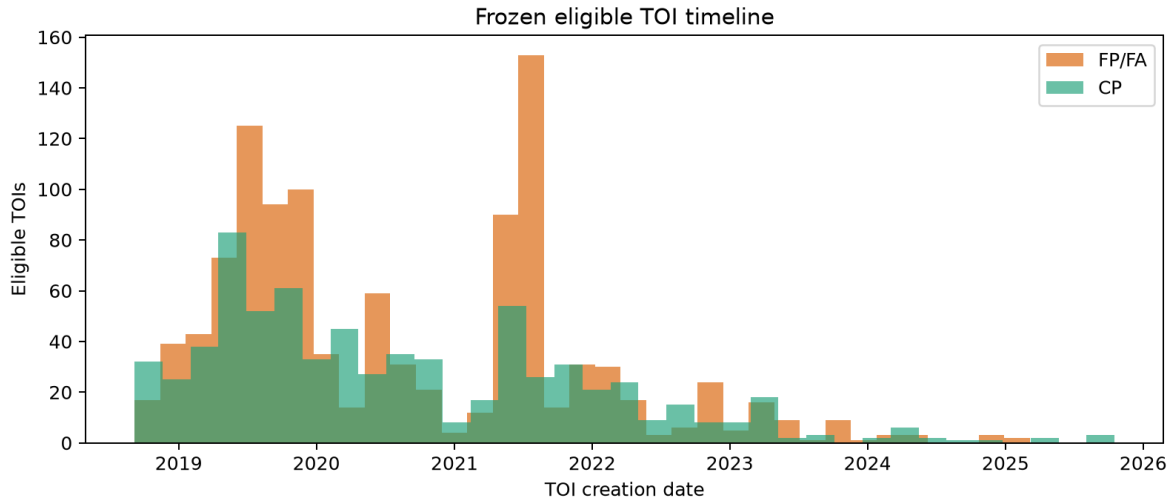


Figure 1. Creation-date distribution of the frozen eligible TOIs. The orange distribution contains FP and FA records, while the green distribution contains CP records. The temporal split uses whole TIC groups rather than individual rows.

4. Methods

4.1 Light-curve preparation and BLS search

For each eligible TIC, the pipeline requested only TESS light curves produced by SPOC at 120-second cadence. Lightkurve supplied the archive search and light-curve access interface (Lightkurve Collaboration et al., 2018). The default SPOC quality mask was applied during acquisition. The local preprocessing code then removed nonfinite points, sorted observations by time, and rejected isolated flux outliers beyond eight robust standard deviations from the median. A 301-point median filter estimated the long-timescale trend. Dividing by that trend and normalizing to a median of one produced the flattened flux used by BLS.

After cleaning, the BLS search used 2,000 trial periods from 0.5 days to the smaller of 30 days or half of the observed time baseline. The duration grid contained five geometrically spaced values constrained by the shortest trial period. The highest-power BLS solution defined the candidate period, epoch, duration, depth, power, signal-to-noise estimate, and phase centre. This search used only the observed time and flux arrays. A code-level leakage check rejected any formal experiment frame containing catalogue period or midpoint fields.

Using the BLS solution, the curve was folded into the phase interval from -0.5 to 0.5. The global view contained 201 mean-flux bins over the full orbit. The local view contained 101 bins centred on the detected event and extended to three candidate durations on either side. Empty phase bins were filled with the curve's median flux. The two fixed-size views allowed each sample to enter the same neural-network architecture even though the original observations had different lengths and sector coverage.

4.2 Baseline and Stage 1 model

The non-neural baseline was a class-balanced logistic regression over BLS and folded-curve diagnostics. Its features included candidate period, duration, depth, BLS power, BLS signal-to-noise estimate, number of observed transits, odd-even depth difference, secondary-eclipse depth, transit symmetry difference, and sector-to-sector depth standard deviation. Missing values were median-imputed, and the features were standardized using statistics learned from the training partition.

Stage 1 was a two-branch one-dimensional CNN inspired by the global/local representation used by Shallue and Vanderburg (2018). One branch processed the 201-bin global view and the other processed the 101-bin local view. Each branch used two convolutional layers with ReLU activations, max pooling after the first convolution, and adaptive average pooling after the second. The branch outputs were concatenated and passed through a 64-unit fully connected layer, dropout of 0.2, and one output logit. The final model contained 38,273 trainable parameters.

Training used binary cross-entropy loss, the Adam optimizer, a learning rate of 0.001, batches of 64, and a maximum of 40 epochs. Early stopping ended training after six epochs without validation improvement. Three deterministic seeds, 4000, 4001, and 4002, were run. Seed 4002 had the highest validation PR-AUC and was selected before test evaluation. Figure 2 shows that training loss decreased for all three seeds while validation PR-AUC varied enough to justify reporting seed selection explicitly.

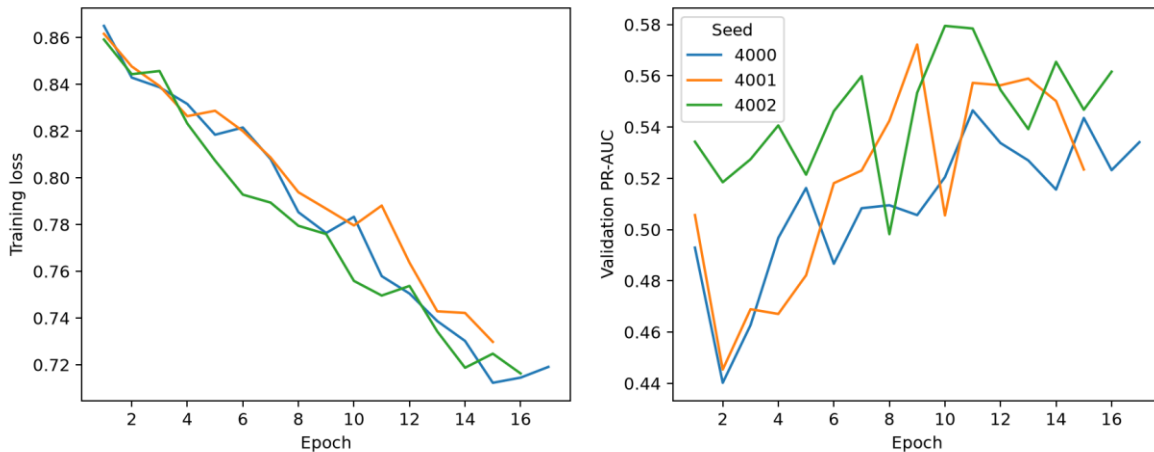


Figure 2. Stage 1 learning curves for seeds 4000, 4001, and 4002. Seed 4002 achieved the highest validation PR-AUC, 0.579, and was selected without reference to the sealed test labels.

4.3 Diagnostic Stage 2

Stage 2 was a second class-balanced logistic regression. It received the Stage 1 score plus the ten diagnostic features used by the baseline. The odd-even feature measured the absolute difference between alternating event depths. The secondary-eclipse feature measured a possible flux decrease near phase 0.5. The symmetry feature compared the mean flux on the two sides of the primary event. Sector consistency was the standard deviation of depth estimates across sectors when more than one usable sector was present.

To avoid training Stage 2 on overly optimistic Stage 1 predictions, training-set CNN scores were generated by three-fold cross-fitting. Whole TIC groups remained together in every fold. For each fold, Stage 1 was trained on the other folds, with a grouped inner validation subset for early stopping, and then scored the held-out fold. Joining the three held-out predictions produced one out-of-fold Stage 1 score for every training record. The final Stage 2 model was fitted on those scores and the diagnostic features. Validation and test records were scored by the selected full-training Stage 1 model.

The full pipeline is summarized in Figure 3.

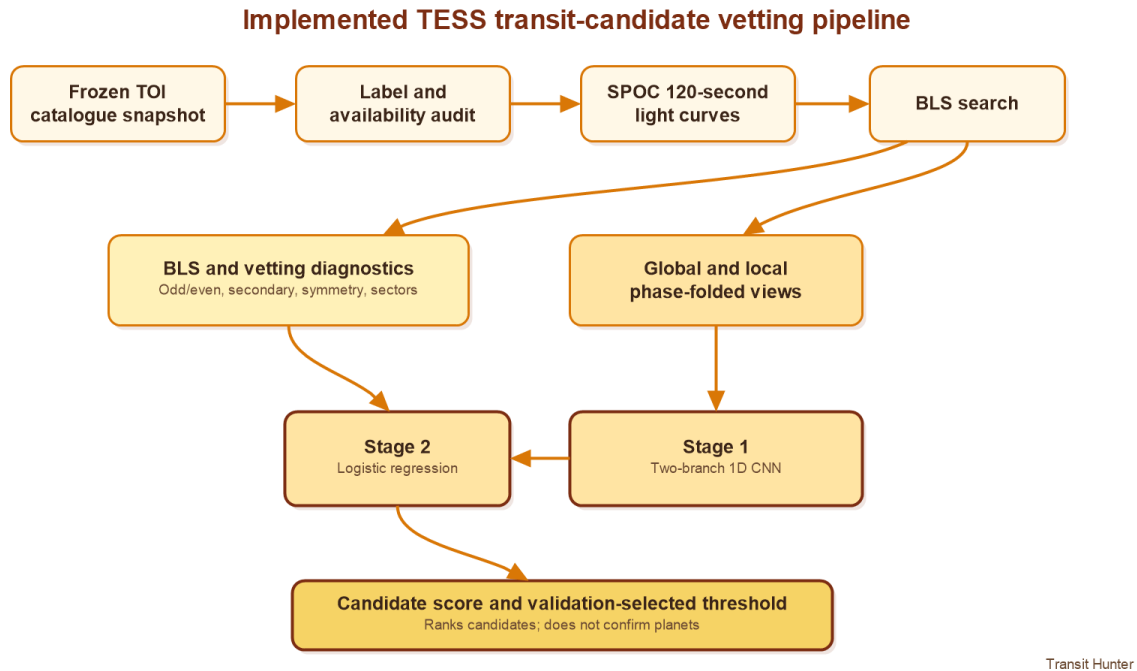


Figure 3. Implemented two-stage pipeline. Catalogue labels define the supervised outcome, but catalogue period and midpoint are not model inputs.

5. Evaluation design

The primary evaluation metric was area under the precision-recall curve. PR-AUC was emphasized because precision directly measures the proportion of accepted candidates that are positive, while recall measures the proportion of positives retained. Precision-recall plots are especially informative when class balance makes accuracy or an ROC curve alone difficult to interpret (Saito & Rehmsmeier, 2015). ROC-AUC was still reported because it summarizes ranking over true-positive and false-positive rates across thresholds.

Alongside those ranking metrics, the report uses precision, recall, F1, specificity, false-positive rate, and the complete confusion matrix at one declared operating point. Accuracy was not used as the main result. A model can achieve a superficially strong accuracy by exploiting the majority class, and accuracy hides whether errors are missed planets or accepted false positives.

For the baseline, Stage 1, and Stage 2, the classification threshold was chosen independently on the validation partition to reach at least 0.90 recall. The selected thresholds were 0.3436, 0.3451,

and 0.3425, respectively. Only after these choices were fixed were the models evaluated on the test set. The design therefore measures how a high-recall operating rule selected on older records transferred to newer labelled TOIs.

Finally, uncertainty in the difference between Stage 2 and Stage 1 false-positive rates was estimated with 2,000 bootstrap resamples of the test set. Resampling was grouped by TIC so that multiple TOIs around one star were not treated as independent observations. The reported interval is the 95% bootstrap interval for Stage 2 FPR minus Stage 1 FPR. Negative values favour Stage 2. This interval measures test-sample uncertainty conditional on the selected fitted models; it does not include variation from retraining the models with different random seeds.

6. Results

6.1 Baseline and ablation comparison

Table 2 compares four saved experiment rows representing three distinct fitted conditions. The BLS/diagnostic baseline and the diagnostics-only Stage 2 row are identical because both use the same diagnostic feature set without a CNN score. Stage 1 produced the best F1 and recall at its validation-selected threshold, but it also produced the highest false-positive rate. Combining the CNN score with the diagnostics produced the best PR-AUC, ROC-AUC, precision, specificity, and false-positive rate. It did not produce the best recall or F1.

Table 2 *Sealed temporal test performance*

Model	ROC-AUC	PR-AUC	Precision	Recall	F1	FPR	TP	FP	TN	FN
BLS/diagnostic baseline	0.688	0.683	0.585	0.809	0.679	0.563	114	81	63	27
Stage 1 global/local CNN	0.689	0.664	0.580	0.851	0.690	0.604	120	87	57	21
Diagnostics only	0.688	0.683	0.585	0.809	0.679	0.563	114	81	63	27
Stage 1 score + diagnostics	0.711	0.719	0.591	0.716	0.647	0.486	101	70	74	40

The ranking curves support the same interpretation. Stage 1 had a test ROC-AUC of 0.689 and PR-AUC of 0.664. Stage 2 increased those values to 0.711 and 0.719. The improvement in PR-AUC means the combined score produced a better precision-recall trade-off across possible thresholds, even though the particular high-recall threshold selected on validation transferred poorly.

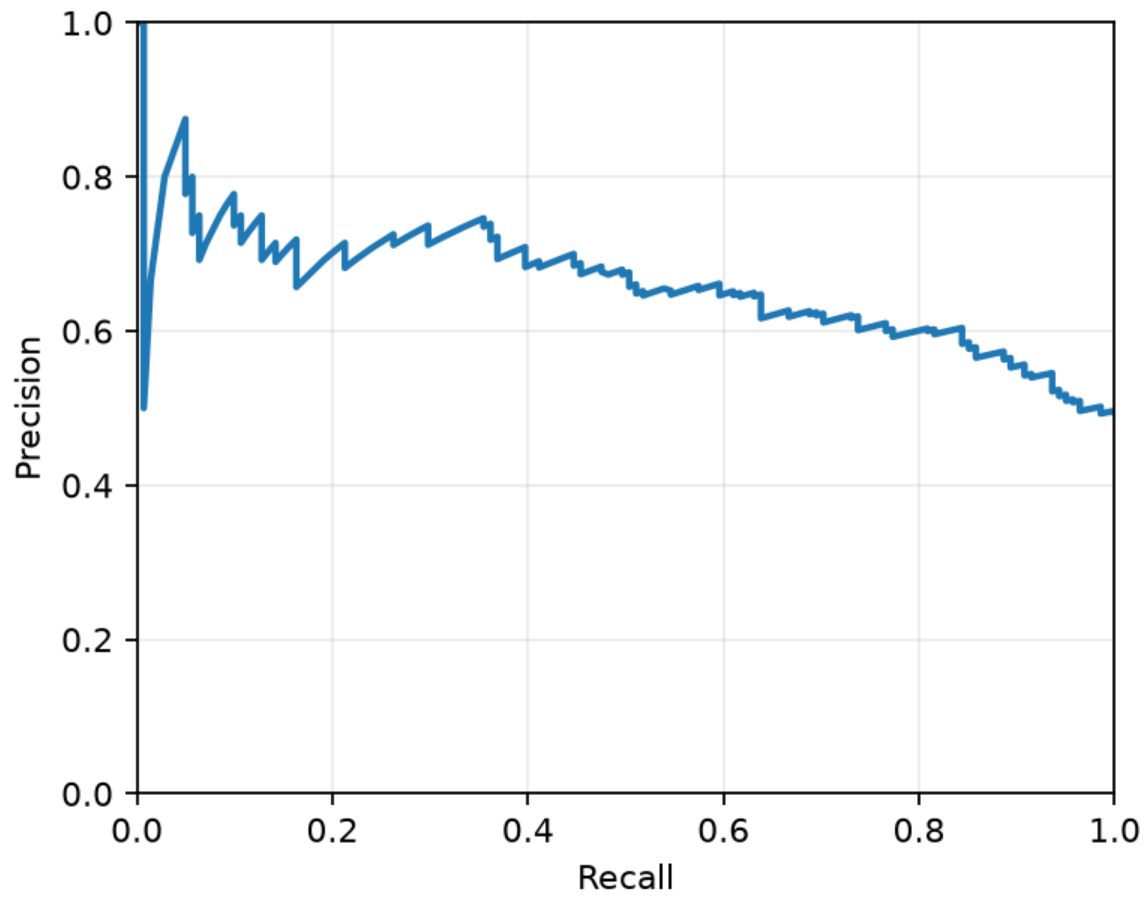


Figure 4a. Stage 1 precision-recall curve on the sealed temporal test set.

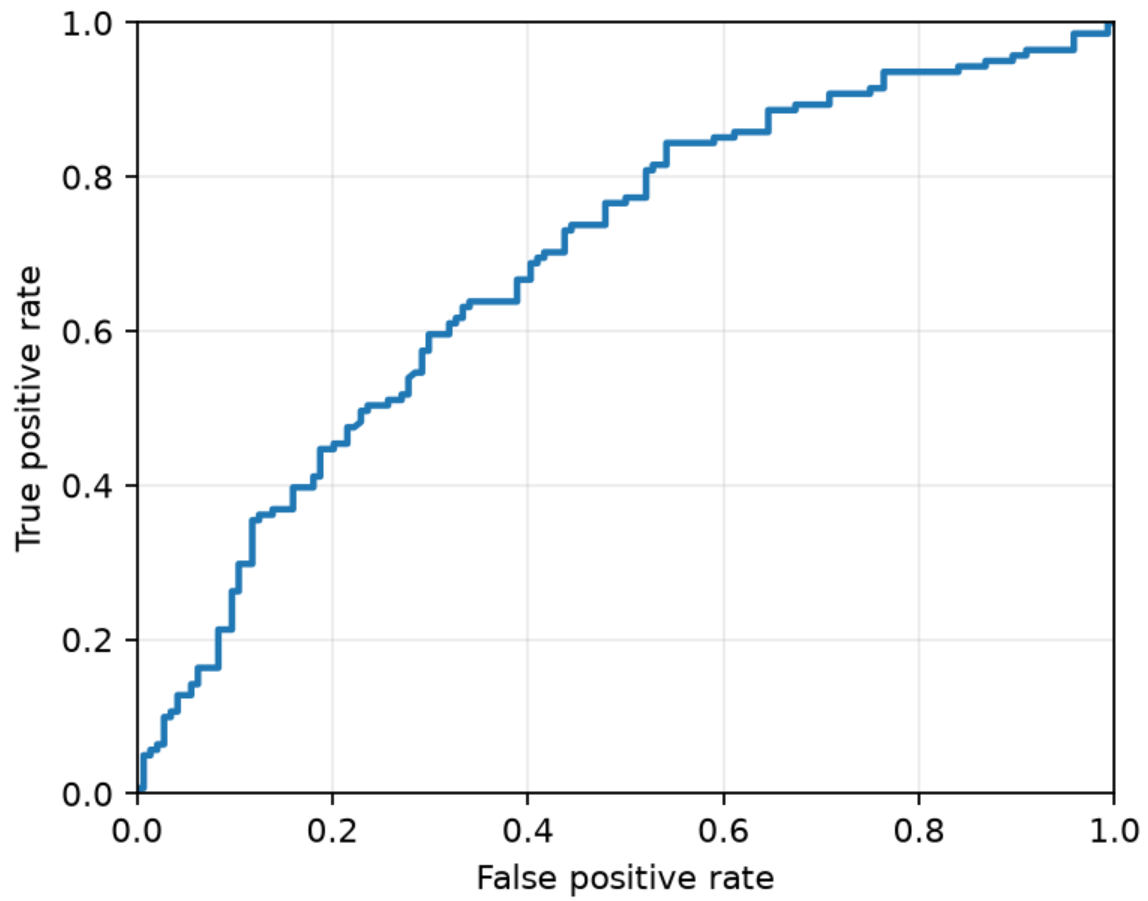


Figure 4b. Stage 1 ROC curve on the sealed temporal test set.

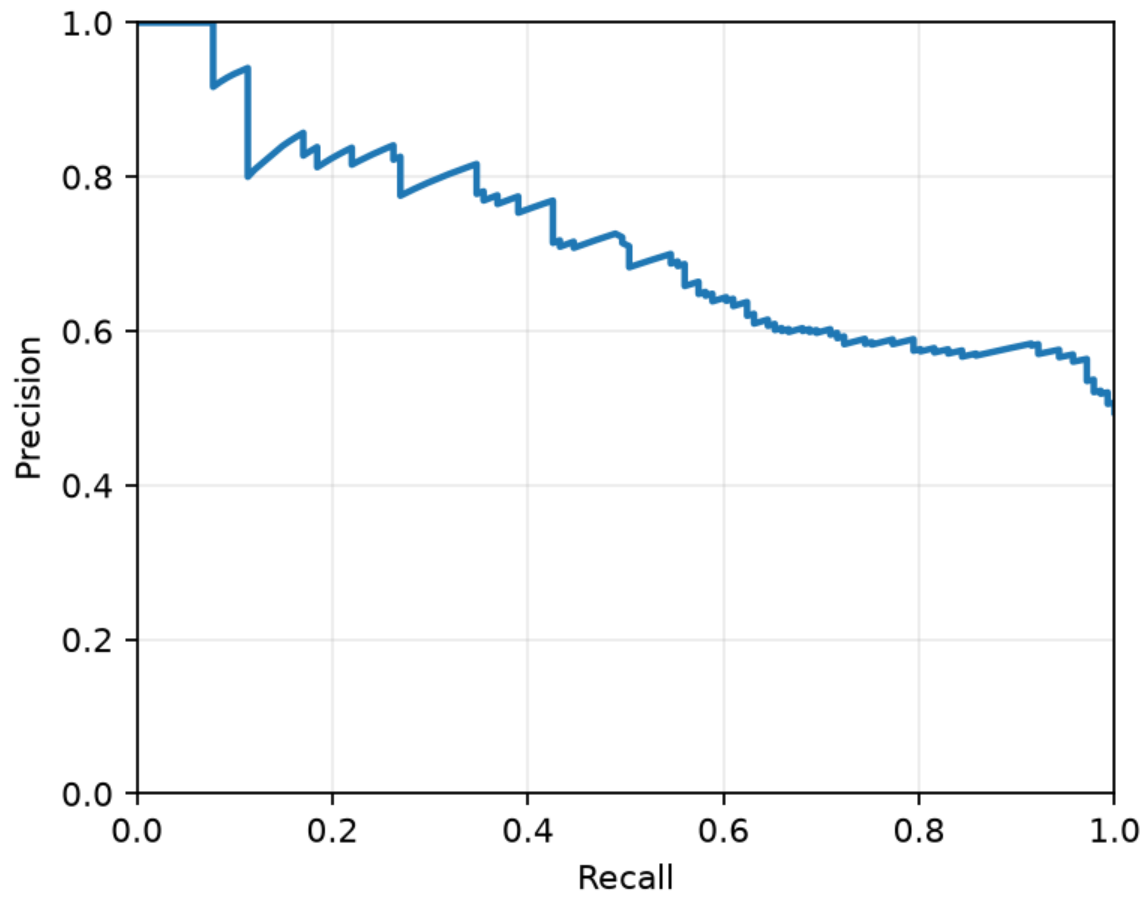


Figure 5a. Stage 2 precision-recall curve on the sealed temporal test set. Stage 2 produced the higher PR-AUC, but thresholded recall must still be considered separately.

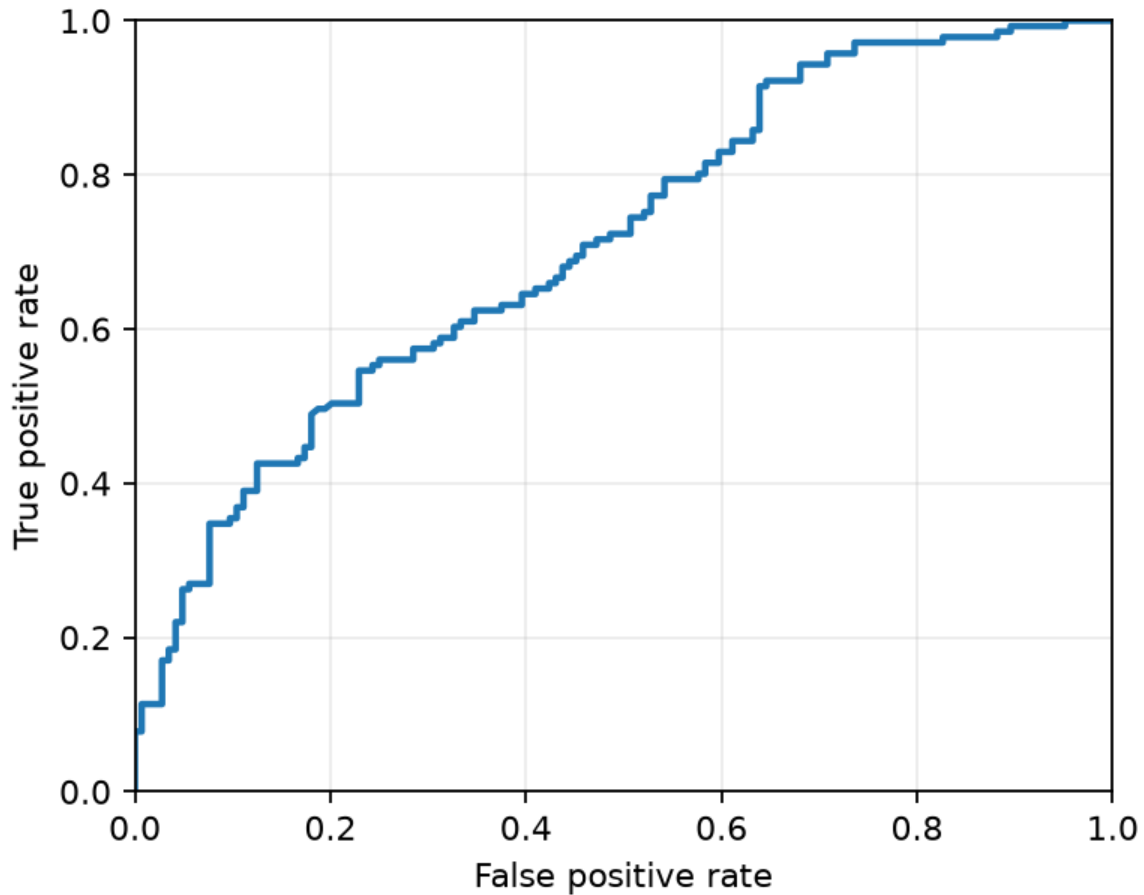


Figure 5b. Stage 2 ROC curve on the sealed temporal test set.

6.2 Operating-point comparison

At the chosen Stage 1 threshold, 120 of 141 confirmed planets were accepted and 21 were missed. Among 144 negative records, 87 were incorrectly accepted and 57 were rejected. This produced recall of 0.851, precision of 0.580, and a false-positive rate of 0.604.

At the chosen Stage 2 threshold, 101 confirmed planets were accepted and 40 were missed. Among the negative records, 70 were accepted and 74 were rejected. Relative to Stage 1, Stage 2 accepted 17 fewer false positives and 19 fewer true positives. Precision rose slightly to 0.591, while recall fell to 0.716. The confusion matrices in Figures 6 and 7 make the net exchange visible. Because the sealed test set is nearly balanced and contains only labelled TOIs, its precision and PR-AUC are conditional on this curated class mixture and should not be transferred directly to an unfiltered TESS population with a different planet prevalence.

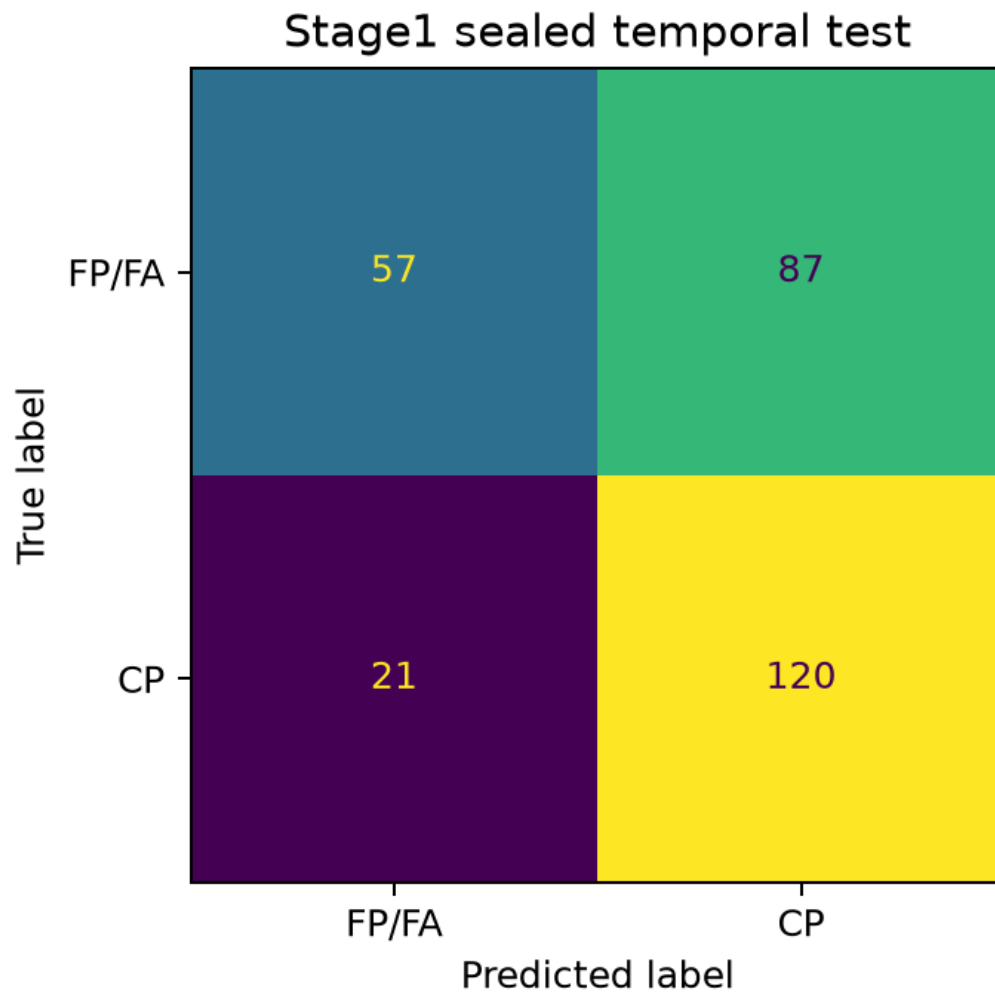


Figure 6. Stage 1 confusion matrix at the validation-selected threshold of 0.3451.

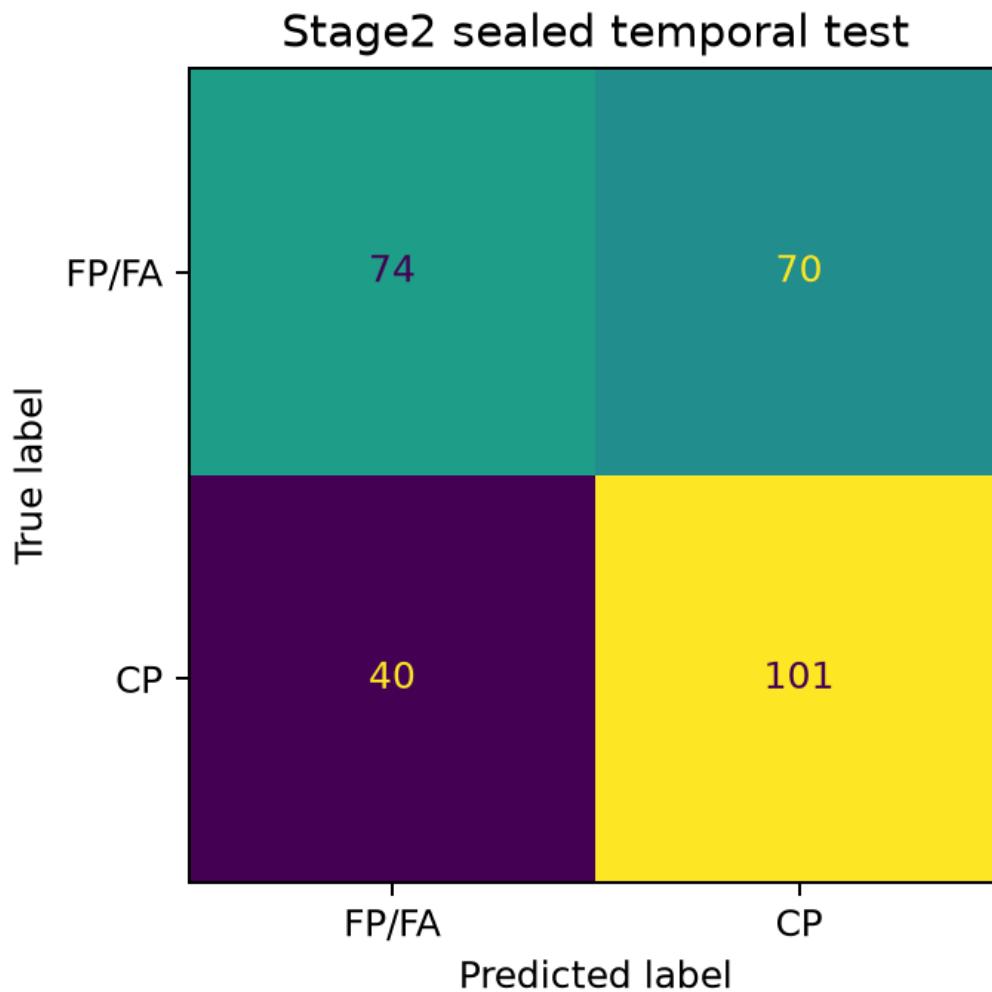


Figure 7. Stage 2 confusion matrix at the validation-selected threshold of 0.3425.

The bootstrap comparison found a Stage 2 minus Stage 1 false-positive-rate difference of -0.118. The TIC-grouped 95% interval was -0.184 to -0.053. Because the full interval is below zero, the reduction in false-positive acceptance was stable across the grouped resamples. The interval does not prove that the same reduction will appear in another TESS population. It quantifies uncertainty within this sealed test population.

6.3 Validation-to-test transfer

Both stages achieved validation recall of 0.903, close to the target of 0.90. Their test behaviour then separated. Stage 1 test recall was 0.851, while Stage 2 test recall fell to 0.716. Stage 2 validation FPR was 0.609 and test FPR was 0.486, so the model became more selective in both directions on the newer data. This is consistent with an operating-point shift rather than a simple improvement in every kind of error.

The descriptive reliability diagram in Figure 8 provides another warning against interpreting the raw scores as calibrated probabilities. Both curves deviate from the diagonal. Because the plot contains only 285 test records, uses quantile bins, and does not show uncertainty intervals, it

should be treated as a qualitative diagnostic rather than a precise comparison of calibration. The scores remain useful for ranking, but a Stage 2 score of 0.70 should not be read as a demonstrated 70% probability that the signal is a confirmed planet.

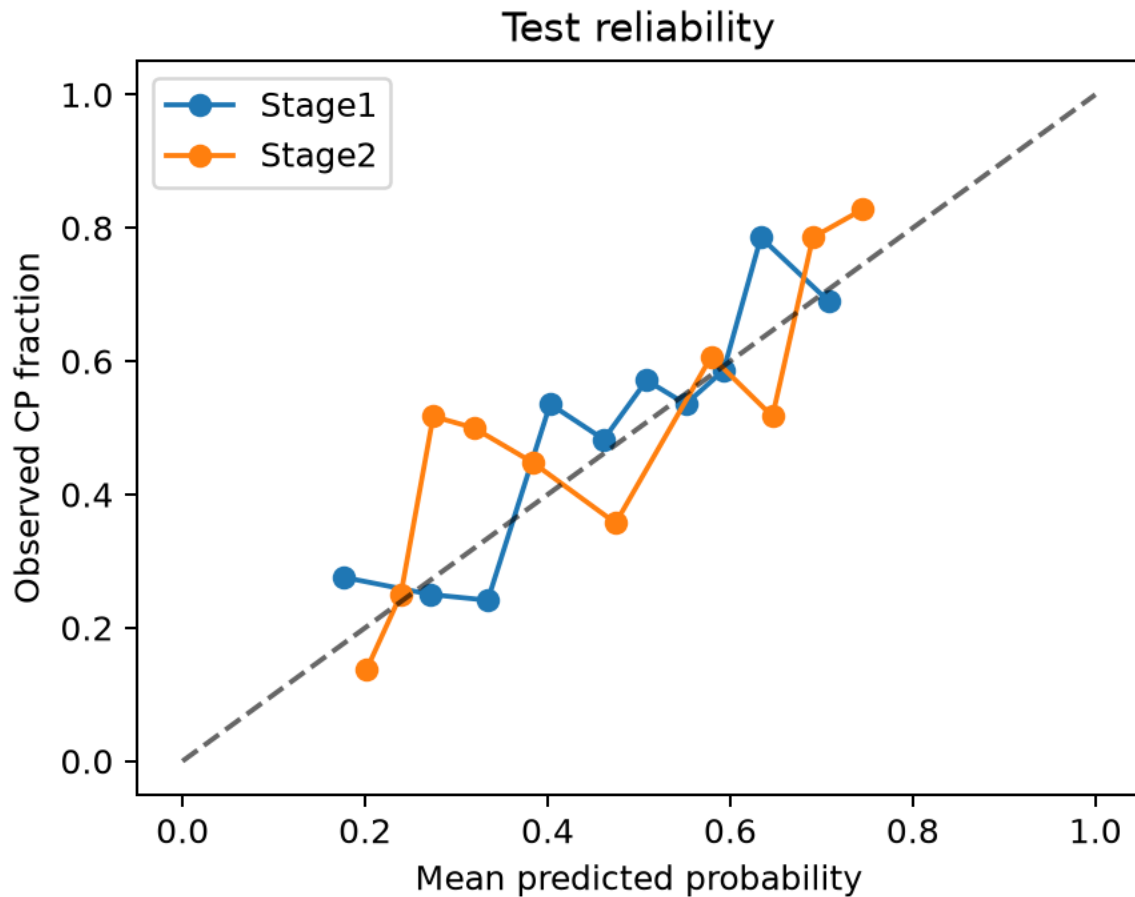


Figure 8. Quantile-binned reliability diagram on the sealed test set. The deviations from the diagonal show that the classification scores are not fully calibrated planet probabilities.

7. Error analysis and limitations

The error gallery in Figure 9 shows the three highest-scoring Stage 2 false positives and the three lowest-scoring false negatives. TOIs 5133.01, 6711.01, and 6693.01 are labelled false positives but received Stage 2 scores from 0.727 to 0.745. Their folded views contain concentrated features near the nominal transit centre, which helps explain why a shape-based model might accept them. TOI 6693.01 also has a broad and irregular local structure that is visibly unlike an ideal shallow transit, showing that a high combined score does not guarantee a physically clean signal.

By contrast, TOIs 5616.01, 6383.01, and 5747.01 are confirmed planets with Stage 2 scores from 0.194 to 0.208. The first has a noisy transit region, the second has a deep narrow feature with substantial local variation, and the third appears to contain more than one depression in its global view. These cases show that confirmed planets do not form one simple shape class after a fixed preprocessing and BLS procedure. They also show why rejecting every low-score case would damage completeness.

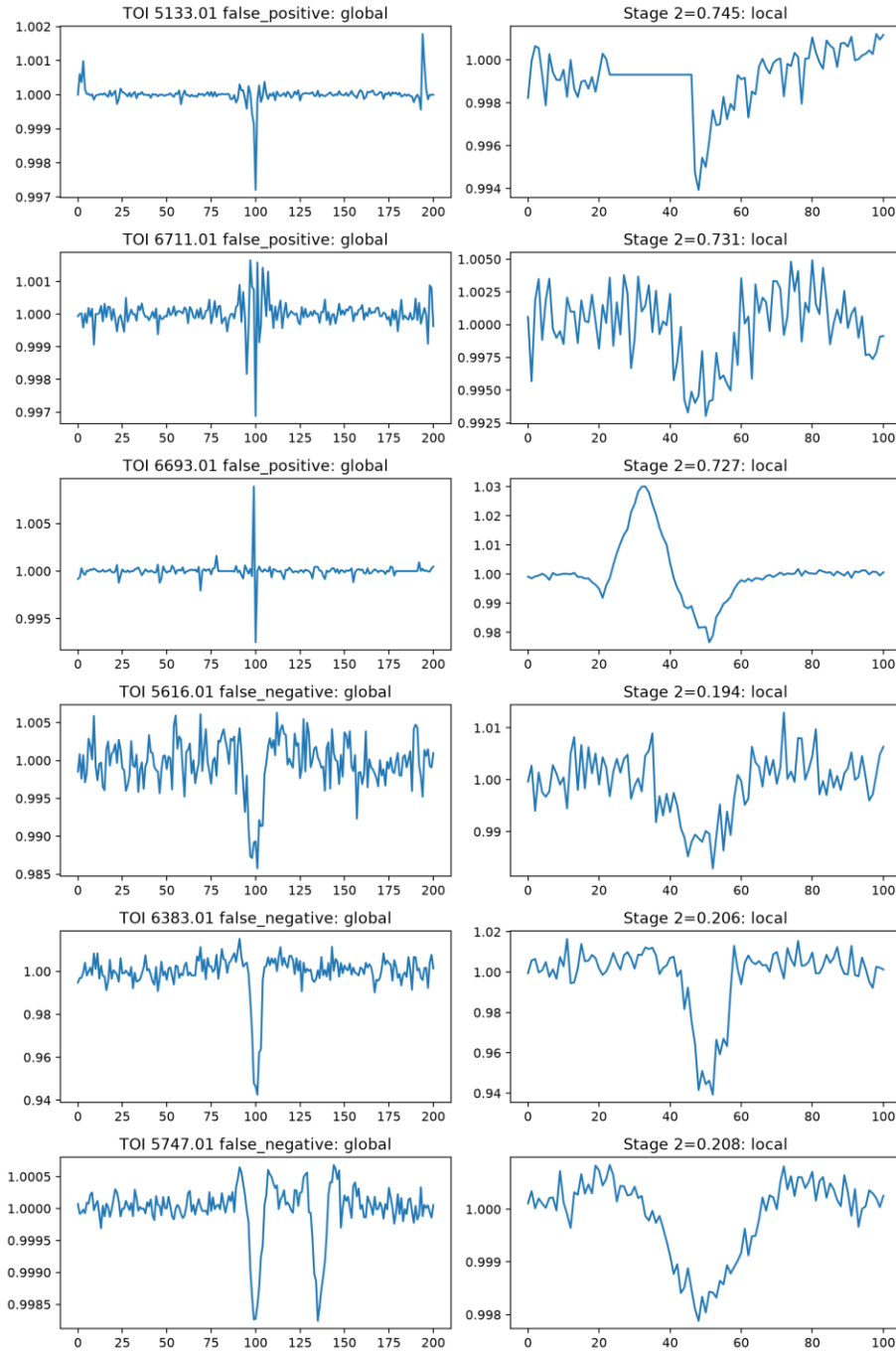


Figure 9. Deterministically selected Stage 2 errors. The first three rows are the highest-scoring false positives; the last three are the lowest-scoring false negatives. These examples are diagnostic illustrations, not a complete taxonomy of failure modes.

One limitation is the label source itself. TFOPWG dispositions summarize real follow-up work, but the catalogue is continually updated and does not represent a random sample of all TESS stars. A signal must first attract attention and receive enough follow-up to acquire a mature label. Brightness, signal strength, observing sector, target priority, and follow-up resources can all shape the labelled population. Freezing the snapshot makes this experiment reproducible, but it does not remove those selection effects.

A second limitation is product availability. The experiment includes only TOIs with SPOC 120-second light curves. The 265 supervised TOIs without matching products were excluded, so the results cannot automatically be generalized to full-frame-image light curves or other cadences. Likewise, each sample used one deterministic BLS search with a fixed 0.5-to-30-day range. Longer-period planets, transit-timing variations, single-transit events, and signals whose strongest BLS peak is an alias may be represented poorly.

A third limitation is the diagnostic set. The implemented second stage used light-curve and sector-level features, but no centroid or pixel-level diagnostics. Centroid motion can help identify signals originating from a nearby blended star rather than the target. ExoMiner and ExoMiner++ use substantially richer inputs, including centroid information and detailed data-validation products (Valizadegan et al., 2022, 2025). The omission kept this project feasible, but it limits what Stage 2 can learn about contamination.

A fourth limitation is temporal transfer. The test set contains newer catalogue groups than training and validation. That is more demanding than a random split, but time is mixed with other changes: later sectors, changing follow-up practices, catalogue revisions, and different class composition. The experiment demonstrates a performance change on newer records; it does not identify which part of that change is causal.

Finally, the model comparison is intentionally narrower than the proposal's full list of possibilities. The repository contains implementations for a compact LSTM and Transformer, but no sealed-run artifacts support reporting their performance. Kepler-to-TESS transfer learning also remained a stretch objective. Omitting unsupported numbers is preferable to filling the report with unverified comparisons. The completed evidence supports the BLS baseline, global/local CNN, diagnostics-only model, and combined Stage 2 model.

8. Astronomical significance and course connection

The immediate result concerns candidate triage, but the broader scientific issue is catalogue quality. Planet occurrence studies attempt to infer an underlying population from the planets a survey detects. That inference depends on completeness, meaning how many real planets are missed, and reliability, meaning how many catalogue entries are genuine. Bryson et al. (2020) showed with Kepler catalogues that occurrence estimates can change when reliability is handled poorly. A vetting threshold that removes false positives while also removing real planets changes both sides of that calculation.

This project illustrates the problem directly. If only the 17 fewer accepted false positives were reported, Stage 2 would look unambiguously better. If only the 19 additional missed confirmed planets were reported, it would look worse. Both are true. The appropriate threshold depends on the use case. A follow-up program with limited telescope time may accept lower recall to obtain a cleaner short list. A population study may prefer greater completeness and correct statistically for contamination. The classifier cannot choose that scientific priority on its own.

That trade-off connects the project to the course's treatment of observational evidence. Week 4 asks how information about distant objects is extracted from light, and Week 6 extends that

question to telescopes, detectors, and time monitoring (Gill, 2026a, 2026b). Here, a time series of photon measurements becomes a folded light curve, a BLS candidate, and finally a classification score. Every step extracts useful structure, but every step can also introduce assumptions or lose information.

The Week 10 exoplanet material adds a second connection. It describes transits as periodic reductions in stellar brightness and explains that detection methods favour some kinds of planets over others (Gill, 2026c). Short-period planets create more repeated events during a limited observing window. Large planets usually create deeper transits. Geometric alignment determines whether a transit occurs from Earth's viewpoint at all. The labelled TOI sample therefore cannot be read as a direct census of every planetary system. It is the outcome of transit geometry, instrument sensitivity, search procedures, and follow-up decisions.

The Week 13 material supplies a final boundary. A potentially habitable planet is not evidence of life, and a transit candidate is not yet a confirmed planet (Gill, 2026d). Better candidate vetting can eventually contribute to more reliable estimates of how common different planetary environments are, but this experiment does not measure habitability or an occurrence rate. Its contribution is smaller and more defensible: it tests one method for changing the balance between false-positive rejection and retained confirmed planets in a defined TESS sample.

9. Conclusion

This project tested whether adding a diagnostic second stage improved false-positive rejection beyond a global/local CNN on frozen, labelled TESS candidates. The hypothesis was partially supported. Stage 2 increased PR-AUC from 0.664 to 0.719 and reduced the false-positive rate from 0.604 to 0.486. The grouped bootstrap interval for the FPR difference remained below zero, and the combined model accepted 17 fewer false positives than Stage 1.

However, Stage 2 also accepted 19 fewer confirmed planets, reducing test recall from 0.851 to 0.716. The high-recall validation threshold did not transfer cleanly to the newer test population. The evidence supports a narrower conclusion: the diagnostics improved ranking and selectivity, but not every operational outcome. Threshold choice remains a scientific decision shaped by the relative cost of wasted follow-up and missed planets.

The experiment also reinforced a broader lesson from PHY4000. Astronomical knowledge is built by extracting physical meaning from limited observations. TESS light curves contain evidence about planetary transits, but the route from a brightness dip to a statement about planetary populations includes instruments, algorithms, labels, and selection effects. Treating those layers explicitly is part of doing the astronomy correctly.

Reproducibility

The public repository is available at <https://github.com/aidan-elliott/transit-hunter>, and a supplementary project website is available at <https://transit-hunter.aidanelliott.com>. The sealed experiment was produced by `scripts/run_real_experiments.py`, and its run manifest recorded repository HEAD `fe45cc4e7b8ceef1d999a06d34a2ea6fe25db43a` together with hashes of the model

configuration and input manifests. Later repository state preserves the executable script and the saved artifacts used in this report; the original manifest did not record whether the working tree contained uncommitted changes. The environment used Astropy 8.0.1, Lightkurve 2.6.0, NumPy 2.4.6, pandas 2.3.3, scikit-learn 1.9.0, and PyTorch 2.13.0. The dataset and split manifests record all 1,812 eligible samples, split counts, zero TIC crossings, and the frozen catalogue checksum. Raw target-specific FITS products are not committed, but the repository includes the catalogue query, label snapshot, eligibility files, split assignments, reproduction commands, model artifacts, metrics, and report figures.

Generative AI use declaration

OpenAI Codex was used as a supplementary tool while preparing this report. It helped organize existing project material, supported the drafting and review process, checked internal consistency, and assisted with editing for clarity. The numerical analysis, models, dataset manifests, and figures were produced through the Transit Hunter project before the report was drafted. All cited sources and reported values were checked against the original publications, official documentation, course materials, or saved project artifacts. The author reviewed and revised the final report and remains responsible for the submitted work.

References

- Bryson, S., Coughlin, J. L., Kunimoto, M., & Mullally, S. E. (2020). Reliability correction is key for robust Kepler occurrence rates. *The Astronomical Journal*, 160(5), 200. <https://doi.org/10.3847/1538-3881/abb316>
- Gill, A. (2026a). *Light and matter: Reading messages from the cosmos* [Course slides]. Algonquin College.
- Gill, A. (2026b). *Telescopes: Portals of discovery* [Course slides]. Algonquin College.
- Gill, A. (2026c). *Other planetary systems: The new science of distant worlds* [Course slides]. Algonquin College.
- Gill, A. (2026d). *Life in the universe* [Course slides]. Algonquin College.
- Guerrero, N. M., Seager, S., Huang, C. X., Vanderburg, A., Garcia Soto, A., Mireles, I., Hesse, K., Fong, W., Glidden, A., Shporer, A., Latham, D. W., Collins, K. A., Quinn, S. N., Burt, J., Dragomir, D., Crossfield, I., Vanderspek, R., Fausnaugh, M., Burke, C. J., ... Winn, J. N. (2021). The TESS Objects of Interest catalog from the TESS prime mission. *The Astrophysical Journal Supplement Series*, 254(2), 39. <https://doi.org/10.3847/1538-4365/abefe1>
- Kovács, G., Zucker, S., & Mazeh, T. (2002). A box-fitting algorithm in the search for periodic transits. *Astronomy & Astrophysics*, 391, 369-377. <https://doi.org/10.1051/0004-6361:20020802>
- Lightkurve Collaboration, Cardoso, J. V. d. M., Hedges, C., Gully-Santiago, M., Saunders, N., Cody, A. M., Barclay, T., Hall, O., Sagar, S., Turtelboom, E., Zhang, J., Tzanidakis, A., Mighell, K., Coughlin, J., Bell, K., Berta-Thompson, Z., Williams, P., Dotson, J., & Barentsen, G. (2018). *Lightkurve: Kepler and TESS time series analysis in Python* [Computer software]. Astrophysics Source Code Library. <https://ui.adsabs.harvard.edu/abs/2018ascl.soft12013L>
- NASA Exoplanet Archive. (2025). *TESS Objects of Interest table data column definitions*. https://exoplanetarchive.ipac.caltech.edu/docs/API_TOI_columns.html
- National Aeronautics and Space Administration. (n.d.). *TESS documentation*. High Energy Astrophysics Science Archive Research Center. Retrieved August 14, 2026, from <https://heasarc.gsfc.nasa.gov/docs/tess/documentation.html>
- Ricker, G. R., Winn, J. N., Vanderspek, R., Latham, D. W., Bakos, G. Á., Bean, J. L., Berta-Thompson, Z. K., Brown, T. M., Buchhave, L., Butler, N. R., Butler, R. P., Chaplin, W. J., Charbonneau, D., Christensen-Dalsgaard, J., Clampin, M., Deming, D., Doty, J., De Lee, N., Dressing, C., ... Villaseñor, J. (2015). Transiting Exoplanet Survey Satellite. *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1), 014003. <https://doi.org/10.1117/1.JATIS.1.1.014003>
- Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. <https://doi.org/10.1371/journal.pone.0118432>

- Shallue, C. J., & Vanderburg, A. (2018). Identifying exoplanets with deep learning: A five-planet resonant chain around Kepler-80 and an eighth planet around Kepler-90. *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>
- Space Telescope Science Institute. (n.d.). *TESS data products*. Mikulski Archive for Space Telescopes. Retrieved August 14, 2026, from <https://archive.stsci.edu/missions-and-data/tess/data-products>
- Sullivan, P. W., Winn, J. N., Berta-Thompson, Z. K., Charbonneau, D., Deming, D., Dressing, C. D., Latham, D. W., Levine, A. M., McCullough, P. R., Morton, T., Ricker, G. R., Vanderspek, R., & Woods, D. (2015). The Transiting Exoplanet Survey Satellite: Simulations of planet detections and astrophysical false positives. *The Astrophysical Journal*, 809(1), 77. <https://doi.org/10.1088/0004-637X/809/1/77>
- Valizadegan, H., Martinho, M. J. S., Wilkens, L. S., Jenkins, J. M., Smith, J. C., Caldwell, D. A., Twicken, J. D., Gerum, P. C. L., Walia, N., Hausknecht, K., Lubin, N. Y., Bryson, S. T., & Oza, N. C. (2022). ExoMiner: A highly accurate and explainable deep learning classifier that validates 301 new exoplanets. *The Astrophysical Journal*, 926(2), 120. <https://doi.org/10.3847/1538-4357/ac4399>
- Valizadegan, H., Martinho, M. J. S., Jenkins, J. M., Twicken, J. D., Caldwell, D. A., Maynard, P., Wei, H., Zhong, W., Yates, C., Donald, S., Collins, K. A., Latham, D., Barkaoui, K., Berlind, P., Calkins, M. L., Carden, K., Chazov, N., Esquerdo, G. A., Guillot, T., ... Wilkin, F. P. (2025). ExoMiner++: Enhanced transit classification and a new vetting catalog for 2-minute TESS data. *The Astronomical Journal*, 170, 287. <https://doi.org/10.3847/1538-3881/ae03a4>
- Yu, L., Vanderburg, A., Huang, C., Shallue, C. J., Crossfield, I. J. M., Gaudi, B. S., Daylan, T., Dattilo, A., Armstrong, D. J., Ricker, G. R., Vanderspek, R. K., Latham, D. W., Seager, S., Dittmann, J., Doty, J. P., Glidden, A., & Quinn, S. N. (2019). Identifying exoplanets with deep learning. III. Automated triage and vetting of TESS candidates. *The Astronomical Journal*, 158(1), 25. <https://doi.org/10.3847/1538-3881/ab21d6>